DHRITI GUPTA

7852821439 dhritigupta2411@gmail.com LinkedIn GitHub

PROFILE SUMMARY

Computer Science undergraduate at JK Lakshmipat University and Visiting Undergraduate Student at IIT Kanpur, focused on software and web development. Experienced in building full-stack applications, REST APIs, document-based RAG systems, and ML workflows using Python, JavaScript, React, and Flask/FastAPI.

EDUCATION

B.Tech in Computer Science and Engineering 2024 – Present
JK Lakshmipat University, Jaipur **CGPA: 8.54**

Visiting Undergraduate Student Jan 2026 – May 2026
Indian Institute of Technology Kanpur (IIT Kanpur)

- Selected for competitive academic exchange program; completed advanced Computer Science coursework.

SKILLS

Programming: C, C++, Python, JavaScript
ML/AI: Scikit-learn, XGBoost, LightGBM, CatBoost, Feature Engineering, Model Evaluation
Web: Flask, FastAPI, React.js, HTML, CSS
Database: SQLite, MongoDB, ChromaDB
Tools: Git, GitHub, VS Code, LangChain, Ollama

WORK EXPERIENCE

Data Science Intern – Veracitiz May 2026 – July 2026

- Built data preprocessing pipelines and performed EDA on structured datasets for ML workflows.
- Applied feature engineering and data preparation techniques to improve model-ready datasets.
- Supported ML model development through structured analysis, evaluation, and experimentation.

PROJECTS

Stroke Risk Prediction System — Full-Stack ML Project

- Built a clinical ML system to predict stroke risk from structured patient health data with severe class imbalance.
- Trained and benchmarked Logistic Regression, Random Forest, XGBoost, LightGBM, and CatBoost models.
- Applied threshold tuning and probability calibration to improve minority-class detection and recall.
- Developed Flask REST APIs and a React dashboard for real-time inference, batch processing, and analytics.
- Implemented SQLite-based storage with downloadable CSV/PDF reports; achieved ROC-AUC of approximately 0.82–0.89.

DocuMind – Full-Stack Retrieval-Augmented Generation (RAG) System

- Built a PDF document assistant enabling natural-language question answering over uploaded documents using RAG.
- Engineered PDF ingestion with PyPDFLoader, 500/100 text chunking, `all-MiniLM-L6-v2` embeddings, and persistent ChromaDB.
- Implemented document-scoped top-3 similarity retrieval using `document_id` metadata filtering to ground responses in relevant context.
- Integrated LangChain with locally hosted Ollama `llama3.1:8b` and developed FastAPI REST APIs with a React interface for upload, chat, history, and document management.

ACHIEVEMENTS & CERTIFICATIONS

- Visiting Undergraduate Student – IIT Kanpur (Jan 2026 – May 2026)
- Vice Chancellor's Honor List – Academic Excellence
- OpenAI x NxtWave Buildathon – Recognized for System Design Innovation
- NPTEL Employment Communication: A Lab-Based Course – Top 2% Rank
- Certifications: Python for Everybody, Exploring C, Generative AI Workshop